The Fundamental Theorem of Calculus

ANSWER KEY

Section 1: FTC Part 1 (Derivative of an Accumulation Function)

1 If $g(x) = \int_1^x \sqrt{t^3 + 1}\, dt$, find $g'(x)$.

a. $\sqrt{x^3 + 1}$ ✓

b. $\frac{3x^2}{2\sqrt{x^3 + 1}}$

c. $\sqrt{x^3 + 1} - \sqrt{2}$

d. $\frac{1}{3}(x^3 + 1)^{3/2}$

By FTC Part 1, $\frac{d}{dx}\left[\int_a^x f(t)\, dt\right] = f(x)$. Thus $g'(x) = \sqrt{x^3 + 1}$.

2 Find $\frac{d}{dx}\left[\int_x^5 \cos(t^2)\, dt\right]$.

a. $-\cos(x^2)$ ✓

b. $\cos(x^2)$

c. $-2x\sin(x^2)$

d. $\cos(25) - \cos(x^2)$

Reverse limits: $\int_x^5 = -\int_5^x$. Differentiating yields $-\cos(x^2)$.

3 Find $h'(x)$ for $h(x) = \int_0^{x^2} \sin(t)\, dt$.

a. $2x\sin(x^2)$ ✓

b. $\sin(x^2)$

c. $2x\cos(x^2)$

d. $\sin(x^4)$

Chain rule with FTC 1: $\frac{d}{dx}\left[\int_a^{u(x)} f(t)\, dt\right] = f(u(x)) \cdot u'(x) = \sin(x^2) \cdot 2x = 2x\sin(x^2)$.

Section 2: FTC Part 1 with Variable Limits on Both Bounds

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4 Find $F'(x)$ for $F(x) = \int_{2x}^{3x} (t^2 + 1)\,dt$.

a. $27x^2 + 3 - 2(4x^2 + 1) = 19x^2 + 1$ ✓

b. $19x^2 + 1$

c. $27x^2 - 8x^2 = 19x^2$

d. $5x^2 + 1$

Split at 0: $\int_{2x}^{3x} = \int_{0}^{3x} - \int_{0}^{2x}$. Differentiating:
$3[(3x)^2 + 1] - 2[(2x)^2 + 1] = 3(9x^2 + 1) - 2(4x^2 + 1) = 27x^2 + 3 - 8x^2 - 2 = 19x^2 + 1$.

5 Evaluate $\frac{d}{dx}\left[\int_{\sin x}^{\cos x} e^t\,dt\right]$.

a. $-e^{\cos x}\sin x - e^{\sin x}\cos x$ ✓

b. $e^{\cos x} - e^{\sin x}$

c. $e^{\cos x}\sin x + e^{\sin x}\cos x$

d. $-e^{\cos x} + e^{\sin x}$

FTC with chain rule: $e^{\cos x} \cdot (-\sin x) - e^{\sin x} \cdot (\cos x) = -e^{\cos x}\sin x - e^{\sin x}\cos x$.

6 Find the inflection points of $g(x) = \int_{0}^{x} (t^3 - 3t)\,dt$.

a. $x = 1$ **and** $x = -1$ ✓

b. $x = 0$ and $x = \pm\sqrt{3}$

c. $x = 3$

d. No inflection points

$g'(x) = x^3 - 3x$. $g''(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \Rightarrow x = \pm 1$. Sign of $g''(x)$ changes at $x = -1$ and $x = 1$.

Section 3: FTC Part 2 (Evaluating Definite Integrals)

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7 Evaluate $\int_1^3 (3x^2 - 4x + 1)\,dx$.

a. 12 ✓

b. 14

c. 10

d. 16

Antiderivative $F(x) = x^3 - 2x^2 + x$. $F(3) = 27 - 18 + 3 = 12$. $F(1) = 1 - 2 + 1 = 0$. Value $= 12 - 0 = 12$.

8 Evaluate $\int_0^{\pi/4} \sec^2(x)\,dx$.

a. 1 ✓

b. $\sqrt{2}$

c. 1/2

d. 0

$[\tan x]_0^{\pi/4} = \tan(\pi/4) - \tan(0) = 1 - 0 = 1$.

9 Evaluate $\int_1^e \frac{4}{x}\,dx$.

a. 4 ✓

b. $4e$

c. $4\ln(e) - 4 = 0$

d. 1

$[4\ln|x|]_1^e = 4\ln(e) - 4\ln(1) = 4(1) - 4(0) = 4$.

Section 4: Average Value of a Function

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10 State the formula for the average value f_{avg} of a continuous function f on $[a, b]$.

a. $f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$ ✓

b. $f_{\text{avg}} = \frac{f(a)+f(b)}{2}$

c. $f_{\text{avg}} = \int_a^b f(x)\,dx$

d. $f_{\text{avg}} = \frac{f(b)-f(a)}{b-a}$

The average value is the total accumulation divided by the width of the interval: $\frac{1}{b-a}\int_a^b f(x)\,dx$.

11 Find the average value of $f(x) = 3x^2 - 2x$ on the interval $[0, 2]$.

a. 2 ✓

b. 4

c. 3

d. 1

$\frac{1}{2-0}\int_0^2 (3x^2 - 2x)\,dx = \frac{1}{2}[x^3 - x^2]_0^2 = \frac{1}{2}[8-4] = \frac{1}{2}(4) = 2.$

12 By the Mean Value Theorem for Integrals, find $c \in [0, 3]$ such that $f(c) = f_{\text{avg}}$ for $f(x) = 2x$.

a. 1.5 ✓

b. 1.0

c. 2.0

d. 0.75

$f_{\text{avg}} = \frac{1}{3}\int_0^3 2x\,dx = \frac{1}{3}[x^2]_0^3 = \frac{1}{3}(9) = 3$. Set $2c = 3 \Rightarrow c = 1.5$.

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